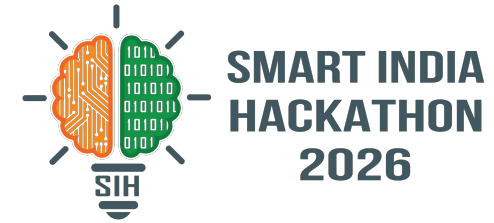


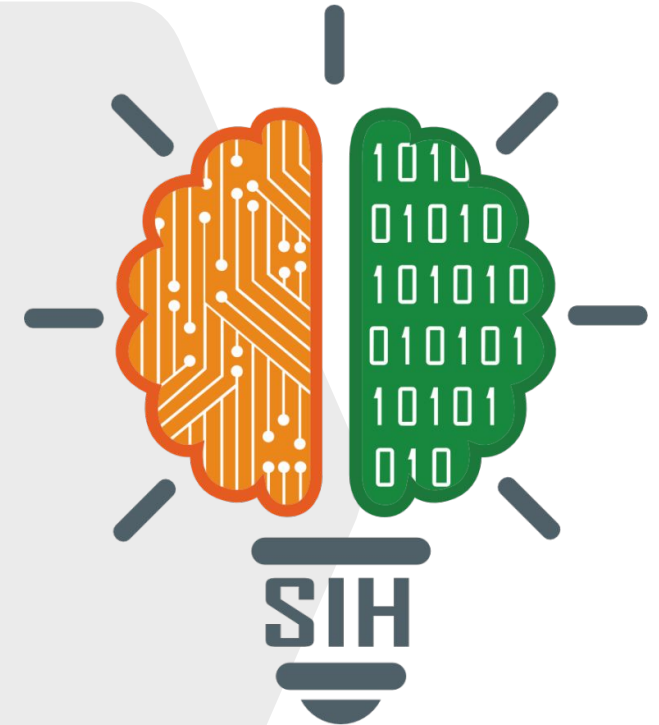
SMART INDIA HACKATHON 2026



3D PROPERTY EXTENSION MAPPING SYSTEM

From Land Records to a Living Digital Property Map.

- **Problem Statement ID – SIH26011**
- **Problem Statement Title- 3D ULPIN Generation
and vertical Property Mapping System**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 133324**
- **Team Name - ZIVRA**



3D PROPERTY EXTENSION ID MAPPING SYSTEM

CORE PROBLEM

Old and scattered land records complicate the process of establishing **property ownership** and **visualizing data** related to land effectively. These challenges result in conflicts, illegal deals, land invasions, and inefficient infrastructure planning. **3D PROPERTY EXTENSION ID** solves this problem by incorporating **property information** into **government programs** using **3D mapping technology**.

SOLUTION

- Links land, property, building, and **public asset data** through a single platform.
- Displays plots, boundaries, apartments, shops and complexes through **precise spatial data**.
- Pinpoints any public land/building impacted by roads, metro railways and other government works.
- The user may use the **3D PROPERTY EXTENSION ID or Property ID** for searching and validating property data.
- Alerts regarding encroachment, dispute and project impacted properties will speed up government decisions.

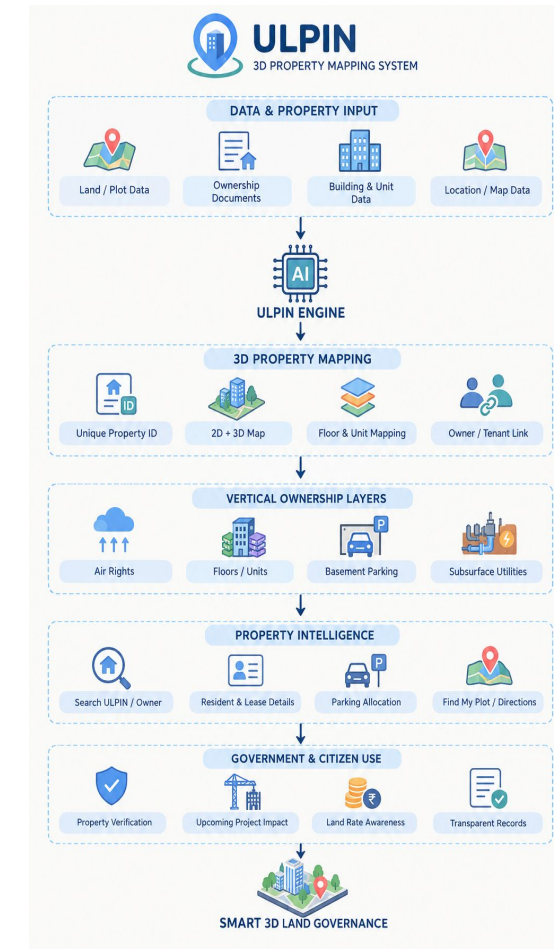
UNIQUE VALUE PROPOSITION (UVP)

3D PROPERTY EXTENSION ID offers a unified, visual and intelligent platform that provides insights into land and property information. It differs from traditional land record systems as it integrates the concepts of **3D PROPERTY EXTENSION ID**, **property information**, and **3D visualization**, **public asset protection**, and **government project impact mapping**.

- View land, buildings, flats, stores and complexes in 3D
- Link **3D PROPERTY EXTENSION ID** with Property ID and other records.
- Recognize government lands and buildings impacted by projects.
- Easily find and validate property information.
- Support decision making of government for risk assessment and infrastructure development.

IMPACT

- Enables **data-driven land** and **infrastructure planning**.
- Improves public access to verified property information.
- Helps authorities **identify risks** and **affected assets quickly**.
- Supports accurate **property identification** and **boundary verification**.
- **Protects public land** and buildings while supporting sustainable development.



1 DATA ACQUISITION & INTEGRATION



Land & Property Data

- ULPIN / Property ID
- Ownership & land records
- Plot boundaries
- Apartments, shops & complexes
- Government asset information



Spatial Data

- GIS / Geospatial data
- Satellite / survey data
- Location & boundary coordinates



Data Integration

2 PROCESSING & 3D MAPPING ENGINE



Data Processing

- Data validation & cleaning
- ULPIN-Property ID linking
- Spatial data processing
- Ownership & property mapping



3D Visualization Engine

- 3D plot & building generation
- Property boundary mapping
- Apartment / shop / complex visualization
- Interactive map layers



Impact Analysis

- Road / metro project mapping
- Government project impact detection
- Encroachment & land-use identification



Interactive Results

3 OUTPUTS & USER INTERFACES



3D Interactive Map

- View plots & boundaries
- Explore buildings & properties
- Search by ULPIN / Property ID



Property Information

- Ownership details
- Property type & location
- Linked land/building records



Project Impact Assessment

- Identify affected properties
- Government infrastructure overlay
- Impact visualization



Government & Public Dashboard

- Government data management
- Public property search
- Planning & decision support

TECH STACK



HTML

Provides the structure of the web application.



CSS

Handles the UI design, layouts, cards, buttons and styling.



JavaScript

Manages application logic, interactions, property data and 3D functions.



Inline SVG

Creates custom 2D/3D-style maps, buildings, plots and visual elements directly in the prototype.



Static File

All code is in one HTML file — no backend, no libraries.

FEASIBILITY



- **Practical Implementation**
Modular architecture allows mapping, property details and analytics to be developed incrementally.
- **Accessible Data**
Works with open GIS data and government portals (e.g., Bhuvan, 3D PROPERTY EXTENSION ID) for mapping and property info.
- **Scalable Infrastructure**
Cloud deployment ensures real-time access, high availability and easy scaling.
- **Operational Feasibility**
A cloud-based system can support centralized access, updates and maintenance.



VIABILITY

- **Real-World Need**
Improves access to reliable land and property information.
- **Multiple Users**
Useful for citizens, buyers, real-estate professionals and government departments.
- **Revenue Potential**
Scope for premium analytics, API access and enterprise services.
- **Future Expansion**
Can integrate ownership, zoning, land-use and additional government services.

IMPACT AND BENEFITS

Impact on Land Transparency



- Visualization of land parcels and their boundaries
- Ensures easy access to property information
- Prevents confusion and lack of information

Impact on Digital Governance



- Integrates 3D PROPERTY EXTENSION ID and visualization of GIS
- Ensures effective administration of land records
- Facilitates data-driven decision-making



Impact on Property Transactions



- Ensures faster validation of property ownership
- Makes it easier for purchasers to make decisions
- Prevents disputes related to land ownership

Impact on Urban & Land Planning



- Facilitates planning of zoning and development
- Ensures visualization of land use and surroundings
- Facilitates infrastructure planning



ULPIN & Land Data References

ULPIN / Bhu-Aadhaar – Department of Land Resources, Government of India

DILRMP – Digital India Land Records Modernization Programme

Geo-referencing of Cadastral Maps – Department of Land Resources

Bhu-Naksha / Digital Cadastral Mapping – NIC / Government initiatives

WHAT WE CREATED

OUR PROTOTYPE

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OUR RESEARCH PAPER

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OUR DEMO VIDEO

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OUR GITHUB

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REFERENCE PAPERS

- [1] 3D Cadastres in India: Examining the Status and Potential for Land Administration and Management in Delhi, *Land Use Policy*, 2020.
- [2] Trends in 3D Cadastre – A Literature Survey, *Land Use Policy*, 2023.
- [3] Design, Development and Usability Testing of an LADM-Compliant 3D Cadastral Prototype System, *Land Use Policy*, 2020.
- [4] Visualization Requirements for 3D Cadastral Systems, *Computers, Environment and Urban Systems*, 2013